NYCMOS
Selectable Homodyne/Sliding IF Downconversion Mixer

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Motivation

- Help students understand/visualize the effect of downconversion by providing a downconversion mixer
- Demonstrate difference between homodyne and heterodyne mixers
- Sliding IF mixer can double as a normal heterodyne mixer as well, so students get exposure to 3 different downconversion approaches
- Expose intermediate outputs by passing them through buffers.
- Exposure to AM demodulation

Design Goals

On Chip

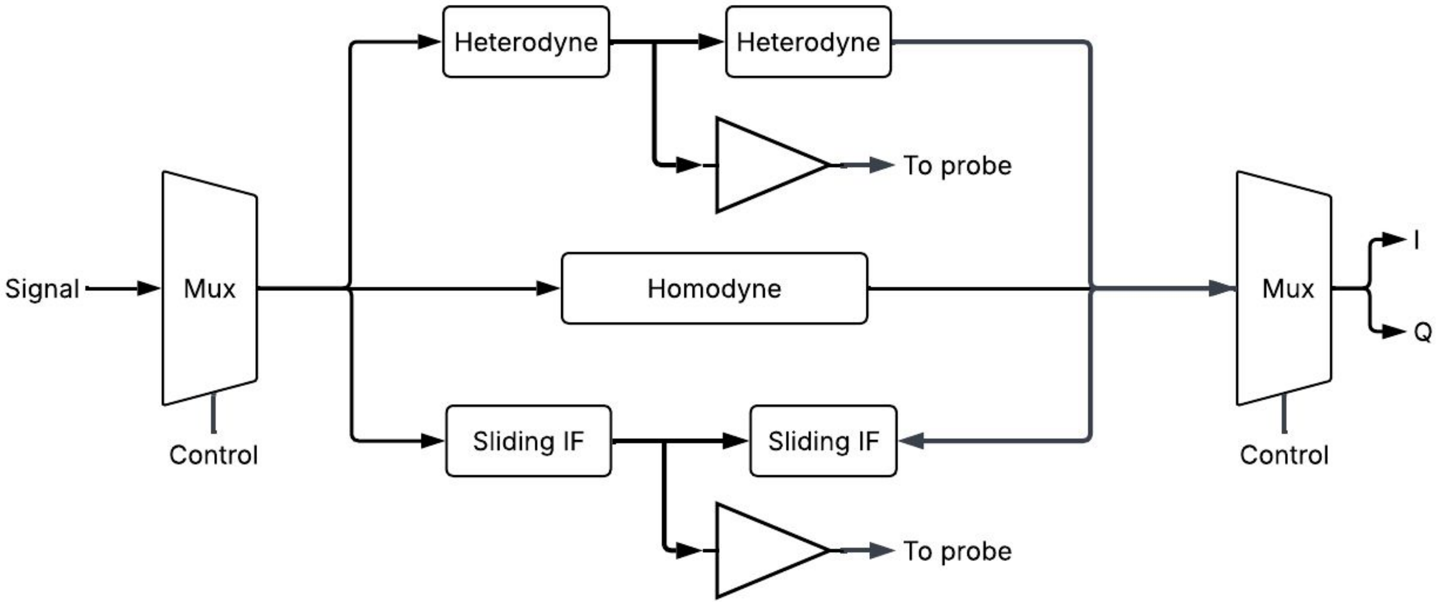
- Homodyne Downconversion Mixer
- Sliding IF Downconversion Mixer
    - Two mixers + 1 divider
- LPF for baseband output
- 4 mux switches to select which mixer is active, and which mixer output goes to the LPF
- 3 switches for picking between internal and external current mirrors for mixers
- 6 buffers to expose internal signals for measurement/observation

Design Goals

Off Chip

- LO
  - Team Prof. Morbius from SSCS Chipathon 2026 used Si535x-B20QFN-EVB
- External Current Mirror
  - From Mosbius chip
- AM signal, generated via signal generator/external chip

Block Diagram



Divider

To Do